Executive Post Graduate Programme in Generative AI and Agentic AI

Course Schedule

Week	Session Title	Date	Time	Instructor
Module 1: Foundation Models & Transformer Architecture (6 Weeks)				
1.1 AI & Deep Learning Essentials		**1 Week**		
1	AI & Deep Learning Essentials	4th April 2026	10 am - 1 pm	Niloy Ganguly
1.2 The Transformer Architecture		**2 Weeks**		
2	Transformer Architecture, Attention, Tokenization	11th April 2026	10 am - 1 pm	Niloy Ganguly
3	Context window, Decoding, Metrics	18th April 2026	10 am - 1 pm	Niloy Ganguly
1.3 Working with Foundation Models		**3 Weeks**		
4	Models and APIs - OpenAI, Anthropic, HuggingFace, structured outputs	25th April 2026	10 am - 1 pm	Sourangshu Bhattacharya
5	Prompting - 0-shot, few-shot, role-prompting, optimization.	2nd May 2026	10 am - 1 pm	Sourangshu Bhattacharya
6	Prompt security, performance evaluation, and budgeting	9th May 2026	10 am - 1 pm	Sourangshu Bhattacharya
Assignment and grading doubt-clearing session		**16th May 2026**	10 am - 1 pm	
Module 2: Advanced Prompting & RAG Systems (6 Weeks)				
2.1 Advanced Prompt Engineering		**2 Weeks**		
7	Advanced Prompting: Chain-of-Thought, Self-consistency, ReAct	23rd May 2026	10 am - 1 pm	Koustav Rudra

8	Advanced Prompting: Prompt optimization, Prompt security, Verbalized Sampling	30th May 2026	10 am - 1 pm	Koustav Rudra
2.2 RAG Fundamentals		**2 Weeks**		
9	RAG: Motivation, architecture, and sparse retrieval	6th June 2026	10 am - 1 pm	Koustav Rudra
10	RAG: Dense Retrieval, Semantic Search	13th June 2026	10 am - 1 pm	Pawan Goyal
2.3 Production-Grade RAG & Agentic Orchestration		**2 Weeks**		
11	RAG for Production: query routing, retrieval gateways	20th June 2026	10 am - 1 pm	Pawan Goyal
12	Agentic Orchestration: Planners vs executors, Context and Memory	27th June 2026	10 am - 1 pm	Pawan Goyal
Test 1 - Proctored (2 hrs) Modules 1 and 2		**4th July 2026**	10 am - 1 pm	
Module 3: LLM Fine-tuning & Alignment (6 Weeks)				
3.1 Fine-Tuning Fundamentals		**2 Weeks**		
13	Decision Frameworks, Fine-tuning Approaches, and Full Fine-Tuning	11th July 2026	10 am - 1 pm	Jiaul Paik
14	Data Preparation and Training Mechanisms for Finetuning	18th July 2026	10 am - 1 pm	Jiaul Paik
3.2 Parameter-Efficient Fine-Tuning		**3 Weeks**		
15	Additive and Soft prompting PEFT	25th July 2026	10 am - 1 pm	Plaban Bhowmik
16	Reparameterization-based PEFT	1st August 2026	10 am - 1 pm	Plaban Bhowmik
17	LLM Alignment: RLHF, DPO	8th August 2026	10 am - 1 pm	Plaban Bhowmik
3.3 Revision and Hands-on		**1 Week**		

18	Revision of fine-tuning concepts and hands-on Small Language models	15th August 2026	10 am - 1 pm	Jiaul Paik
Assignment and grading doubt-clearing session		**22nd August 2026**	10 am - 1 pm	
Module 4: Multimodal & Agentic AI (6 Weeks)				
4.1 Vision-Language Models & Introduction to Image Generation		**3 Weeks**		
19	VLM Architectures	29th August 2026	10 am - 1 pm	Somak Aditya
20	Applications and Multimodal Prompting	5th Sept 2026	10 am - 1 pm	Somak Aditya
21	Multimodal RAG	12th Sept 2026	10 am - 1 pm	Somak Aditya
4.2 Agentic AI Systems		**3 Weeks**		
22	RAG to Assistants and Agents	19th Sept 2026	10 am - 1 pm	Debaditya Roy
23	Tool Use and Function Calling	26th Sept 2026	10 am - 1 pm	Debaditya Roy
24	LangChain and LangGraph	3rd Oct 2026	10 am - 1 pm	Debaditya Roy
Test 2 - Proctored (2 hrs) Modules - 3 and 4		**17th Oct 2026***	10 am - 1 pm	
Module 5: Production Deployment & Industry Capstone (6 Weeks)				
5.1 Model Serving & API Development		**2 Weeks**		
27	REST APIs with Fast API, Real-time vs. Batch Inference	10th Oct 2026	10 am - 1 pm	Abhijnan Chakraborty
	Announcement of Capstone Challenge	17th Oct 2026 [After the test]		
28	Model Context Protocol, Orchestration Frameworks	24th Oct 2026	10 am - 1 pm	Abhijnan Chakraborty
5.2 Responsible and Trusted GenAI		**2 Weeks**		

29	Responsible & Trusted AI Essentials	31st Oct 2026	10 am - 1 pm	Sundar Narayanan (AI Tech Ethics)
	Diwali	8th Nov 2026		
31	Public Case Studies on Implementations & Pitfalls to Avoid	21st Nov 2026	10 am - 1 pm	Sundar Narayanan (AI Tech Ethics)
5.3 Industry Capstone Project		**2 Weeks**		
30	Feedback on Capstone Challenge Problems and Solution Ideas	14th Nov 2026	10 am - 1 pm	Abhijnan Chakraborty/ Sundar Narayanan
32	Evaluation of Capstone Challenge	28th Nov 2026	10 am - 1 pm	TBD
Top 10 teams come to IIT Kharagpur to present their capstone projects		**5th Dec 2026**		

*Durga Puja Shashthi is on 17th Oct